GSK-3α regulates production of Alzheimer's disease amyloid-β peptides

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Alzheimer's disease is associated with increased production and aggregation of amyloid-β (Aβ) peptides[1]. Aβ peptides are derived from the amyloid precursor protein (APP) by sequential proteolysis, catalysed by the aspartyl protease BACE[2], followed by presenilin-dependent γ-secretase cleavage[3]. Presenilin interacts with nicastrin[4,5], APH-1 and PEN-2 (ref. 6), all of which are required for γ-secretase function. Presenilins also interact with α-catenin, β-catenin[7,8] and glycogen synthase kinase-3β (GSK-3β)[9–11], but a functional role for these proteins in γ-secretase activity has not been established. Here we show that therapeutic concentrations of lithium, a GSK-3 inhibitor[12], block the production of Aβ peptides by interfering with APP cleavage at the γ-secretase step, but do not inhibit Notch processing. Importantly, lithium also blocks the accumulation of Aβ peptides in the brains of mice that overproduce APP. The target of lithium in this setting is GSK-3α, which is required for maximal processing of APP. Since GSK-3 also phosphorylates tau protein, the principal component of neurofibrillary tangles, inhibition of GSK-3α offers a new approach to reduce the formation of both amyloid plaques and neurofibrillary tangles, two pathological hallmarks of Alzheimer's disease.

Amyloid plaques are composed primarily of 40- and 42-amino-acid peptides—$A\beta_{40}$ and $A\beta_{42}$, respectively—derived from APP[1]. To investigate the role of GSK-3 in the production of $A\beta_{40}$ and $A\beta_{42}$, we treated Chinese hamster ovary cells that stably expressed APP (CHO-APP_{695} cells) with lithium chloride, a direct inhibitor of GSK-3α and GSK-3β[13], for 24 h, and then measured $A\beta_{40}$ and $A\beta_{42}$ secretion by sandwich ELISA (enzyme-linked immunosorbent assay). LiCl inhibited the production of both $A\beta_{40}$ and $A\beta_{42}$ with an IC_{50} (half-maximal inhibitory concentration) of 1–2 mM—within the therapeutic range for lithium—whereas sodium chloride had no effect (Fig. 1a), consistent with a recent report using transient overexpression of the APP carboxy terminus C100 in COS7 cells[14]. The reduction of Aβ was confirmed by metabolic labelling of CHO-APP_{695} cells with ^{35}S-methionine in the presence or absence of LiCl. Aβ peptides could be immunoprecipitated from the medium of control cells, but treatment with LiCl or the γ-secretase inhibitor DFK-167 (ref. 15) markedly reduced the levels of secreted Aβ peptides (Fig. 1b).

Lithium did not affect steady-state levels of APP (Fig. 1c) or the levels of N- and C-terminal fragments of presenilin 1 (PS1), and it did not interfere with the detection of Aβ peptides (data not shown). Lithium therefore appeared to reduce the level of Aβ peptides at a post-translational step, such as APP processing or Aβ stability. To test this, we measured the accumulation of APP processing intermediates in the presence of LiCl. α-Secretase or β-secretase cleavage of APP generates APP C-terminal fragments (CTFs), which are then cleaved by γ-secretase. If γ-secretase is inhibited—for example, with DFK-167—then APP CTFs accumulate[16]. Exposure of CHO-APP_{695} cells to 5 mM lithium caused the accumulation of APP CTFs (Fig. 1c), similar to treatment with DFK-167. The accumulation of the APP CTFs shows that lithium inhibits APP processing at or before the γ-secretase cleavage step.

γ-Secretase is also required for the release of the Notch intracellular domain (NICD)[15]. However, lithium did not inhibit Notch processing, as assessed by immunoblotting with an antibody specific for cleaved Notch (Fig. 1d). ΔE-V1744K, a mutated form of Notch that is not cleaved by γ-secretase[17], was not detected with this antibody (Fig. 1d, lane 3). Thus, lithium is not a general inhibitor of γ-secretase under these conditions, and might

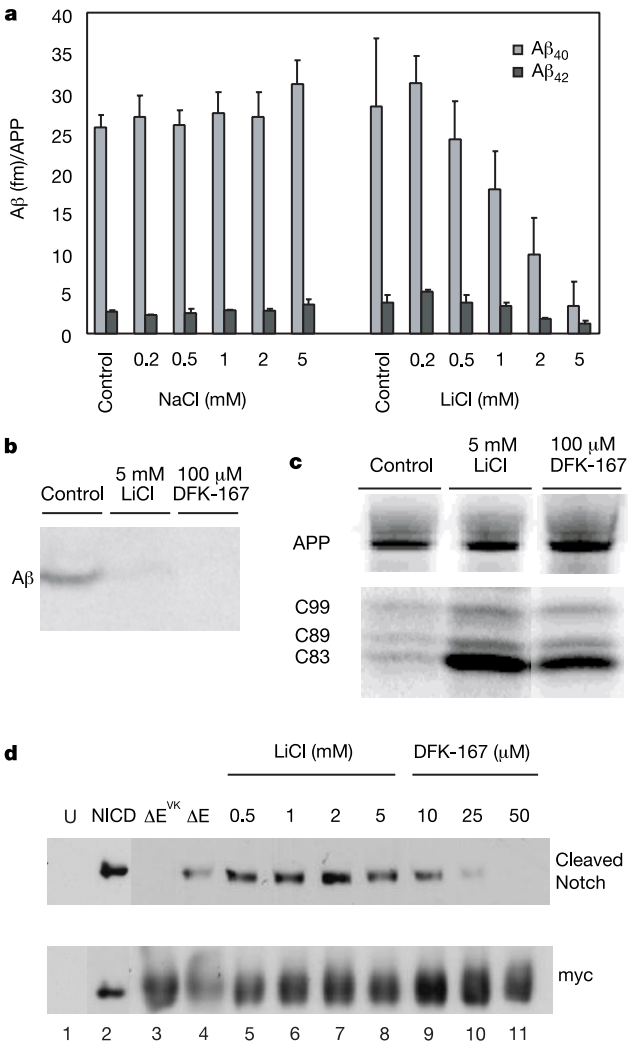

Figure 1 Lithium reduces secreted $A\beta_{40}$ and $A\beta_{42}$, and causes accumulation of APP C-terminal fragments. **a**, CHO-APP_{695} cells were treated with NaCl or LiCl for 24 h. $A\beta_{40}$ (light bars) and $A\beta_{42}$ (dark bars) levels (in femtomoles) were measured by ELISA and normalized to intracellular full-length APP (determined by quantitative immunoblots). Error bars represent standard deviation. **b**, Immunoprecipitated Aβ peptides secreted from CHO-APP_{695} cells after 24 h in culture with either 5 mM LiCl (lane 2) or 100 μM DFK-167 (lane 3). **c**, CHO-APP_{695} cells were treated with 5 mM LiCl or 100 μM DFK-167 for 24 h. Intracellular APP and C-terminal fragments (C99, C89 and C83) were immunoprecipitated, separated by SDS–PAGE, and visualized by PhosphorImager. **d**, CHO-APP_{695} cells were transfected with ΔE-Notch, which is constitutively cleaved by γ-secretase[17] (lanes 4–11), treated with LiCl or DFK-167 for 24 h, and analysed by immunoblotting with an antibody to processed Notch. The NICD is a positive control for the antibody (lane 2); the ΔE^{VK} form of Notch is not cleaved by γ-secretase (lane 3). All Notch constructs have C-terminal myc tags. Overexpression of GSK-3α or GSK-3β had no effect on Notch cleavage (data not shown). U, untransfected.

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specifically affect γ-secretase activity towards APP, or alter access of APP to the γ-secretase complex.

Although lithium is a direct, selective inhibitor of GSK-3 (ref. 13), it also inhibits inositol monophosphatase, related phosphomonoesterases, and phosphoglucomutase[12]. To test the hypothesis that lithium reduces Aβ production through inhibition of GSK-3, we treated CHO-APP_{695} cells with the structurally unrelated GSK-3 inhibitor kenpaullone[18]. Kenpaullone markedly reduced both $A\beta_{40}$ and $A\beta_{42}$ (Fig. 2a), with 50% reduction at 2 μM and >90% reduction at 5 μM. Kenpaullone also inhibits cyclin-dependent kinases (cdks); although this action requires at least 20-fold higher concentrations *in vitro* than are needed for the inhibition of GSK-3 (ref. 18), we nevertheless examined the effect of roscovitine, a cdk inhibitor that does not inhibit GSK-3 (ref. 18), on CHO-APP_{695} cells. Roscovitine had no effect on $A\beta_{40}$ and $A\beta_{42}$ (Fig. 2a), showing that kenpaullone does not inhibit Aβ production through the inhibition of cdks. Both kenpaullone and lithium inhibited GSK-3 in these cells, as indicated by accumulation of β-catenin protein (Fig. 2b) and activation of a β-catenin/Tcf-dependent reporter (Fig. 2c).

PS1 also interacts with β-catenin, and regulates β-catenin protein levels and subcellular localization[10,11]. Because GSK-3 inhibitors cause accumulation of β-catenin[12], it is possible that β-catenin has a role as a downstream effector of PS1 or as a direct modulator of PS1 function. Therefore, we tested whether increasing the levels of β-catenin protein could mimic the effect of lithium and kenpaullone on Aβ production. Overexpression of β-catenin did not affect Aβ production (Fig. 2d) under conditions that activated the β-catenin-dependent reporter (data not shown); infection with semliki forest virus (SFV) encoding β-catenin also had no effect on Aβ production (data not shown). These observations show that an elevated level of β-catenin is not sufficient to cause the decrease in Aβ seen with GSK-3 inhibitors.

These data show that two structurally unrelated inhibitors of GSK-3 reduce the production of Aβ peptides. Although the most parsimonious explanation is that these inhibitors act through the inhibition of GSK-3, the possibility remains that these two agents fortuitously inhibit distinct, as yet unknown targets involved in APP processing. To confirm that GSK-3 regulates APP processing, we reduced the expression of endogenous GSK-3 using RNA interference (RNAi)[19]. In CHO-APP_{695} cells, transfection of short interfering RNAs (siRNAs) directed against GSK-3α and GSK-3β reduced GSK-3 protein levels in an isoform-specific manner, whereas a control siRNA had no effect on GSK-3 expression (Fig. 3a). Selective reduction of GSK-3α protein expression decreased $A\beta_{40}$ and $A\beta_{42}$ levels by 45% and 43%, respectively (Fig. 3b). These data show that GSK-3α is required for maximal production of $A\beta_{40}$ and $A\beta_{42}$, and suggest that GSK-3β can antagonize APP processing in this setting. By contrast, reduction of GSK-3β protein levels resulted in a modest increase in $A\beta_{40}$ and $A\beta_{42}$ concentration; this is surprising, as GSK-3α and GSK-3β are 97% identical within the kinase domains and are apparently redundant in the Wnt pathway, although not in the regulation of NF-κB (nuclear factor-κB)[20]. However, the amino- and carboxy-terminal sequences of GSK-3α and GSK-3β are dissimilar, and this divergence may account for functional differences in the regulation of APP processing.

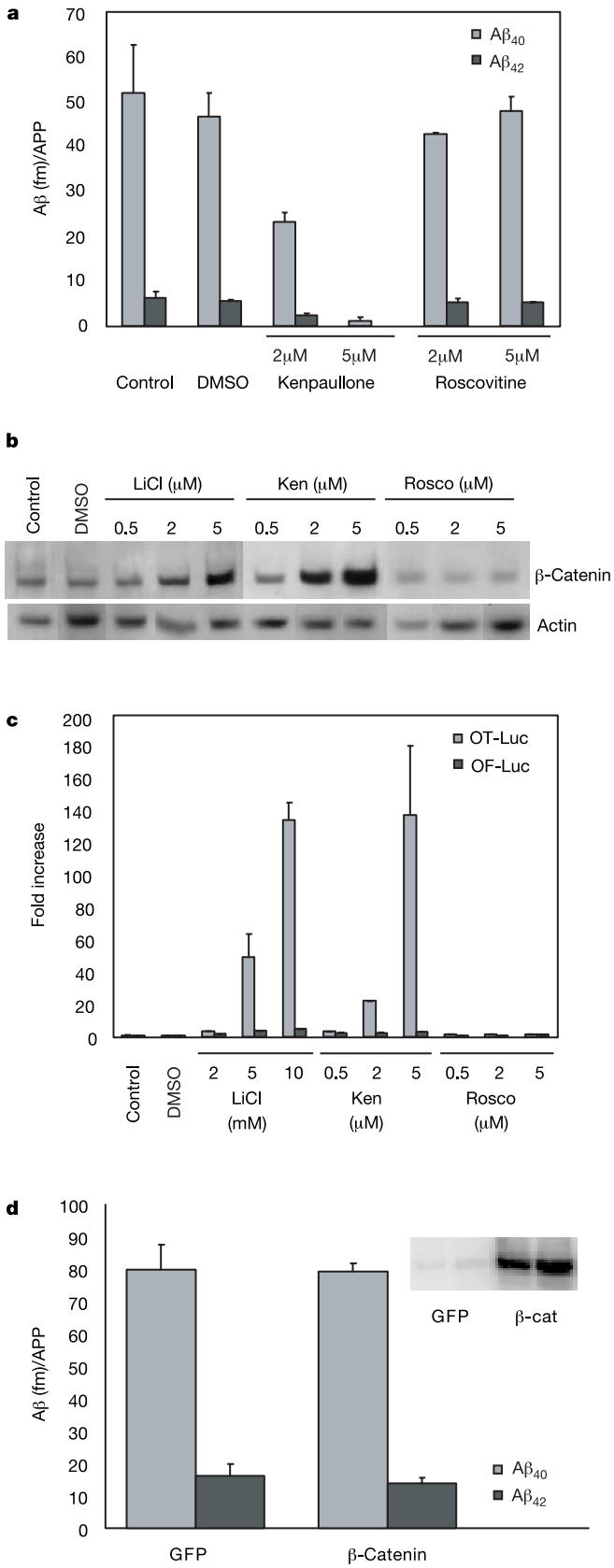

Figure 2 GSK-3 inhibitors reduce Aβ independently of β-catenin stabilization. **a**, CHO-APP_{695} cells were treated with kenpaullone or roscovitine (cdk inhibitor) for 24 h. Secreted $A\beta_{40}$ and $A\beta_{42}$ were measured as in Fig. 1. **b**, β-Catenin levels were assessed by immunoblot in CHO-APP_{695} cells treated with kenpaullone, lithium or roscovitine. Immunoblot for actin is a loading control. Control, untreated cells; DMSO, dimethylsulphoxide; Ken, kenpaullone; Rosco, roscovitine. **c**, CHO-APP_{695} cells transfected with the β-catenin-responsive reporter OT-luciferase (OT-Luc) or the mutated reporter OF-luciferase (OF-Luc) were treated with LiCl, kenpaullone or roscovitine for 24 h, and luciferase activity was measured. **d**, CHO-APP_{695} cells were transfected with green fluorescent protein (GFP; control) or β-catenin. Secreted $A\beta_{40}$ and $A\beta_{42}$ levels were assessed as in Fig. 1. Inset shows immunoblot of endogenous (GFP) and overexpressed (β-cat) β-catenin in duplicate lanes.

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Because either lithium or GSK-3α depletion reduces Aβ levels, we predicted that raising GSK-3α levels would enhance Aβ production.

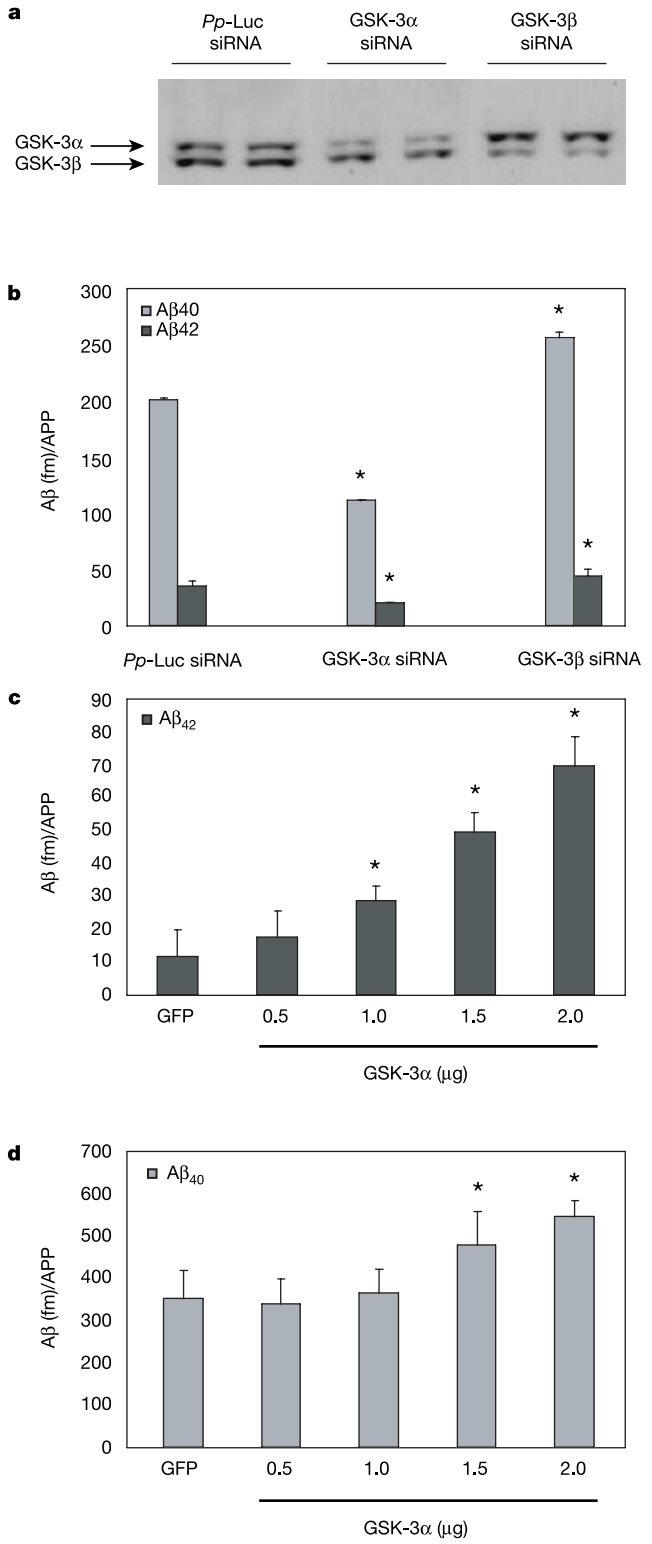

Figure 3 GSK-3α is required for Aβ production. **a**, GSK-3α and GSK-3β levels were measured by immunoblotting in CHO-APP$_{695}$ cells transfected with siRNAs. siRNA against *Pp*-Luc represents control; GSK-3α siRNA selectively reduces GSK-3α protein and GSK-3β-directed siRNA selectively reduces GSK-3β. **b**, Secreted Aβ levels in siRNA transfected cells. The experiment was repeated six times with similar results. **c**, **d**, CHO-APP$_{695}$ cells were transfected with GFP or GSK-3α, and secreted Aβ$_{42}$ (**c**) and Aβ$_{40}$ (**d**) levels were assessed as in Fig. 1. Error bars represent s.e.m. In panels **b**–**d**, asterisks indicate a significant difference from control, as determined by one-way ANOVA ($P < 0.05$).

Overexpression of GSK-3α in CHO-APP$_{695}$ cells increased Aβ$_{40}$ and Aβ$_{42}$ levels in a dose-dependent manner (Fig. 3c, d). Thus, Aβ production correlated with GSK-3α expression using both loss-of-function and overexpression approaches. The isoform-specific depletion of GSK-3 by RNAi, together with the overexpression data, shows that GSK-3α is required for maximal APP processing.

To show that lithium reduces Aβ production in neurons, primary cultures of mouse embryonic cortical neurons were infected with

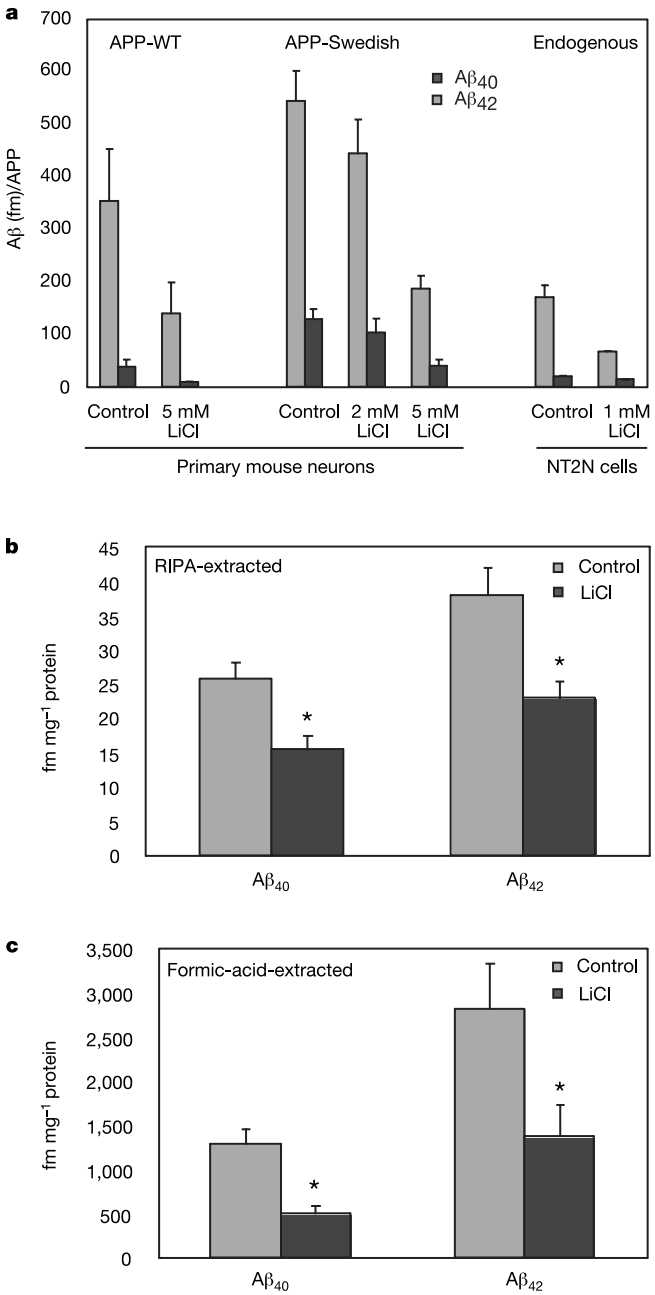

Figure 4 Lithium blocks Aβ accumulation in cultured neurons and in the brains of mice overproducing Aβ peptides. **a**, Embryonic cortical neurons were infected with SFV containing wild-type APP (APP-WT) or APP-Swedish (KM670/671NL), then treated with LiCl for 24 h. Secreted Aβ was measured as described above. **b**, **c**, Three-month-old female mice heterozygous for the APP-Swedish transgene (Tg2576) and the PS1^{P264L} knock-in received 300 mg kg^{-1} LiCl ($n = 7$) or NaCl ($n = 7$) by gastric gavage once daily for 3 weeks, and cortical Aβ accumulation was assessed in RIPA-extracted (soluble) and formic-acid-extracted (insoluble) fractions. Serum lithium levels were 0.8–1.2 mM. Error bars represent s.e.m. Asterisks in panel **b** and **c** denote a significant difference from NaCl-treated animals, assessed by one-way ANOVA ($P < 0.05$).

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SFV encoding wild-type APP (APP_{695}) or the pathogenic APP-Swedish mutation (KM670/671NL), then treated with lithium for 24 h. Production of $A\beta_{40}$ and $A\beta_{42}$ was reduced by 60% and 78%, respectively, for wild-type APP, and to a similar extent for the APP-Swedish mutation (Fig. 4a). A clinically relevant concentration of lithium (1 mM) also reduced the accumulation of endogenous Aβ by 60% after 3 days of culture in the neuronal NT2N cell line (Fig. 4a).

Lithium also reduces Aβ production in an animal model of Alzheimer's disease. Mice expressing pathogenic, familial Alzheimer's disease (FAD)-associated forms of APP (APP-Swedish/Tg2576) and PS1 ($PS1^{P264L/wt}$; ref 21) were treated with daily lithium gavage for 3 weeks, and the levels of Aβ peptides were measured in the brain. Under these conditions, serum lithium levels were 0.8–1.2 mM, within the therapeutic range for patients treated with lithium. In the lithium-treated group ($n = 7$), soluble, RIPA-extracted $A\beta_{40}$ and $A\beta_{42}$ levels were each reduced by 40% (Fig. 4b). Furthermore, insoluble $A\beta_{40}$ and $A\beta_{42}$ extractable with formic acid were reduced by 62% and 51%, respectively (Fig. 4c), showing that a therapeutic dose of lithium markedly reduces the tissue level of Aβ peptides.

In addition to presenilin, nicastrin[4], APH-1 and PEN-2 (ref. 6) are also required for γ-secretase activity, and loss of any of these components affects both APP and Notch processing. By contrast, lithium does not inhibit Notch processing, indicating that lithium is not a general inhibitor of γ-secretase. Although consensus sites for GSK-3 phosphorylation have been identified in PS1 that are important for PS1 stability, mutation of these sites does not affect γ-secretase activity[22]. Thus, it is unlikely that GSK-3α has a role in the biogenesis, stability, or overall activity of the γ-secretase complex. Rather, GSK-3α might regulate γ-secretase activity toward specific substrates, or access of these substrates to the γ-secretase complex. Certain non-steroidal anti-inflammatory drugs (NSAIDs) also reduce $A\beta_{42}$ levels without affecting Notch cleavage[23]. However, the mechanisms of lithium and NSAIDs appear to differ, as NSAIDs shift APP cleavage products from $A\beta_{42}$ to $A\beta_{38}$, whereas lithium reduces overall Aβ production. This apparent difference in mechanism suggests that combination therapy with lithium and NSAIDs could have an enhanced effect in reducing Aβ peptide accumulation.

In summary, we have shown that GSK-3α facilitates APP processing and that lithium inhibits the generation of Aβ peptides through inhibition of GSK-3α. In support of this conclusion: (1) lithium reduces Aβ production in cultured cells and in the brains of mice that overproduce Aβ peptides; (2) kenpaullone, an alternative GSK-3α inhibitor, also inhibits Aβ production; (3) RNAi-mediated depletion of GSK-3α reduces Aβ production; and (4) moderate overexpression of GSK-3α increases Aβ production. Lithium inhibits the GSK-3-mediated phosphorylation of tau[12], which, in its hyperphosphorylated state, is the main component of neurofibrillary tangles[24]. Thus, GSK-3α offers an attractive target for pharmacological agents aimed at reducing the formation of amyloid plaques and neurofibrillary tangles, the pathological hallmarks of Alzheimer's disease. Lithium also protects neurons from pro-apoptotic stimuli and could therefore reduce neuronal cell death associated with Alzheimer's disease[24]. Lithium has been used for more than 50 years to treat bipolar disorder, but has a narrow therapeutic window and a higher frequency of side effects in older patients. Thus, although lithium might be considered for the prevention of Alzheimer's disease, especially in younger patients with FAD mutations or Down's syndrome, new agents that specifically target GSK-3α may prove to be valuable in the treatment of Alzheimer's disease. □

Methods

Plasmids

Human GSK-3α in pMT2 was provided by J. Woodgett. *Xenopus* β-catenin (obtained from B. Gumbiner) was subcloned from pCS2 + into pSFV. Notch-ΔE in pCS2MT and NICD in pCS2MT were provided by R. Kopan. Notch-ΔE-V1744K in pCS2MT was created by site-directed mutagenesis using the QuikChange mutagenesis kit (Stratagene) and confirmed by sequencing. Wild-type APP and APP-Swedish (KM670/671NL) in pSFV were as described previously[25]. OT-luciferase (OT-Luc) was provided by K. Kinzler and B Vogelstein.

Cell culture

CHO_{695} cells (obtained from S. Sisodia) were maintained in MEMα + 5% fetal bovine serum (FBS) with added penicillin/streptomycin and glutamine. To generate NT2N neurons, a human embryonic carcinoma cell line (NTera2/D1 or NT2) was grown and maintained as described[26]. For murine primary embryonic neurons, cortices from embryonic-day-15 mouse brains were isolated and incubated in 0.1% trypsin, HBSS and 0.5 mM EDTA without Ca^{2+} or Mg^{2+} (Invitrogen), and the cells mechanically dissociated using a fire-polished pipette. Cells were plated in DMEM + 10% FBS in poly-D-lysine-coated six-well plates at a density of 10^6 cells per well. Twenty-four hours after plating, the medium was replaced with DMEM + B27 (Invitrogen) to promote neuronal survival and to inhibit the growth of non-neuronal cells. Neurons were used for experiments after three days in culture.

Aβ quantification

CHO-APP_{695} cells were plated in six-well dishes at a density of 5×10^5 cells per well. Stocks of LiCl were prepared in sterile water, whereas DFK-167 (Enzyme Systems Products), kenpaullone (Calbiochem) and roscovitine (Calbiochem) were prepared in dimethylsulphoxide. Drugs were added to cells in fresh medium, and media and cells were collected 24 h later. Aβ determinations from media were made by sandwich ELISA[27]. APP levels were quantified by SDS–polyacrylamide gel electrophoresis (SDS–PAGE) followed by immunoblotting with the N-terminal APP antibody Karen[28], and visualized using ^{125}I-labelled secondary antibodies by PhosphorImager analysis and ImageQuant software (Molecular Dynamics).

Visualization of metabolically labelled APP fragments

CHO_{695} cells plated on six-well dishes were methionine deprived for 30 min by incubation in methionine-free DMEM (Life Technologies) before adding 500 μCi ^{35}S-methionine (Perkin Elmer) per ml of DMEM with 5% dialysed FBS (Invitrogen) and 40 mg ml^{-1} L-proline for 2.5 h. Media was collected and cells (two wells per sample) were rinsed twice with PBS and scraped into RIPA buffer. Lysates were sonicated and centrifuged at 100,000*g* for 20 min. Media was immunoprecipitated with the antibody BAN-50, and cell lysates were immunoprecipitated with the antibody 2493 using protein A/G agarose beads (Santa Cruz Biotechnologies). Proteins were resolved by electrophoresis on Tris-tricine polyacrylamide gels with a 10/16% step gradient. Gels were fixed in 50% methanol + 5% glycerol, dried and exposed to a PhosphorImager screen.

Viral infections

SFV vectors were used to overexpress β-catenin and APP. SFV was prepared and titred in BHK cells, as previously described[29]. Cells were infected with SFV in serum-free DMEM at a multiplicity of infection of 10. One hour after infection, the medium was replaced with MEMα + 5% FBS (CHO_{695} cells) or DMEM + B27 supplement (murine primary neurons).

siRNA preparation and transfection

RNA oligonucleotides were synthesized by Dharmacon. Sequences used were: pGL3-luciferase (*Pp*-Luc) sense, 5′-CUUACGCUGAGUACUUCGAdTdT-3′; *Pp*-Luc antisense, 5′-UCGAAGUACUCAGCGUAAGdTdT-3′; GSK-3β sense, 5′-AUCUUUGGAGCCACUGAUUdTdT-3′; GSK-3β antisense, 5′-AAUCAGUGGCUCCAAAGAUdTdT-3′; GSK-3α sense, 5′-UUCUACUCCAGUGGUGAGAdTdT-3′; GSK-3α antisense, 5′-UCUCACCACUGGAGUAGAAdTdT-3′. Double-stranded RNA was produced using conditions described in ref. 9. siRNA was transfected into CHO-APP_{695} cells plated in six-well dishes using GenePORTER (Gene Therapy Systems). siRNA (25 nM) was co-transfected with 2 μg plasmid DNA and cells were harvested 48 h later. GSK-3α and GSK-3β protein levels were examined using an antibody that recognizes both GSK-3 isoforms (Calbiochem).

Notch processing

CHO-APP_{695} cells were plated at a density of 1×10^5 cells per well in six-well dishes. Cells were transfected with 2 μg of either Notch-ICV or Notch-ΔE. Twenty-four hours later, the medium was changed and drugs were added. After a further 24 hours, cells were harvested as described above, and immunoblotted with both myc (9E10) antibody or cleaved Notch antibody (Cell Signaling Technologies).

Lithium treatment in mice

Transgenic mice expressing APP-Swedish (Tg2576) were crossed to mice carrying a knock-in of the $PS1^{P264L}$ mutation, as described[20], and 3-month-old female Tg2576/PS-$1^{P264L/WT}$ mice ($n = 7$) were administered LiCl or NaCl (300 mg kg^{-1} in 0.4 ml water, by gastric gavage daily for 3 weeks). Animals were allowed free access to food and water, and lithium-treated mice were given free access to a 450-mM NaCl solution. Animals were sacrificed 3 h after the final dose, and Aβ levels were measured after stepwise extraction of Aβ from

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brain tissues. Briefly, cerebral cortices were lysed in RIPA buffer; after centrifugation, insoluble material was extracted in formic acid, as described elsewhere[30]. RIPA- and formic-acid-extracted samples were diluted, and Aβ levels were measured by sandwich ELISA, as above.

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Caenorhabditis elegans early embryogenesis and vulval morphogenesis require chondroitin biosynthesis

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Defects in glycosaminoglycan biosynthesis disrupt animal development and can cause human disease[1–4]. So far much of the focus on glycosaminoglycans has been on heparan sulphate. Mutations in eight *squashed vulva* (*sqv*) genes in *Caenorhabditis elegans* cause defects in cytokinesis during embryogenesis and in vulval morphogenesis during postembryonic development[5,6]. Seven of the eight *sqv* genes have been shown to control the biosynthesis of the glycosaminoglycans chondroitin and heparan sulphate[6–11]. Here we present the molecular identification and characterization of the eighth gene, *sqv-5*. This gene encodes a bifunctional glycosyltransferase that is probably localized to the Golgi apparatus and is responsible for the biosynthesis of chondroitin but not heparan sulphate. Our findings show that chondroitin is crucial for both cytokinesis and morphogenesis during *C. elegans* development.

Glycosaminoglycans (GAGs) or mucopolysaccharides have been of great interest to biologists for decades[2–4,12–15]. Analyses of mutations that cause developmental defects in *Drosophila melanogaster* have shown that GAG biosynthesis, in particular the synthesis of heparan sulphate (HS), is important for intercellular signalling mediated by the Wingless, Hedgehog and fibroblast growth factor pathways[2]. In addition, mutations in GAG biosynthesis have been implicated in human diseases, including a progeroid variant of the connective tissue disorder Ehlers–Danlos syndrome[1] and hereditary multiple exostoses[3], which is characterized by inappropriate chondrocyte proliferation and bone growth. Much of the focus on GAGs has been on HS, in part because of the many ligands that it binds, its action in growth factor signalling, and its role in *Drosophila* development[2,14]. Studies of chondroitin sulphate (CS), another chief class of GAGs in vertebrates, have focused on the development of cartilage, tendon and bone[4].

The *C. elegans* genes *sqv-1* to *sqv-8* are important for both embryonic development and postembryonic vulval morphogenesis[5]. The progeny of mutants homozygous for strong loss-of-

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